

Geographical Variances

It is sometimes asserted that geographical processes operate at different scales [10, 13]. This concept is often dichotomized by terms as "site and situation," "strategical and tactical," "local and global." At a more advanced level, the Fourier interpretation of scale is the spatial wavelength and this provides an operational definition for the study of processes in terms of scale, now generally known as *spectral analysis*. The spectrum is a measure of the proportion of a process which occurs at any one of a large number of scales. A process is thus not asserted to be effective at only one scale unless theoretical or empirical evidence indicates this to be the case. To give a specific example which will be familiar to readers of this journal, a devotee of spectral analysis would assert that quadrat analysis [1, 9, 12] should be undertaken for all possible quadrat sizes, and the results plotted as a function of scale (quadrat size). The resulting curve is the spectrum; if it has distinct peaks then, and only then, are these the appropriate scales at which to examine the process.

The foregoing is by way of an introduction since spectral analysis is a well documented field [4, 5, 19, 25], and the literature includes geographical examples [23, 24]. Our objective is to suggest, introduce, illustrate, and in-

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interpret an analogous method of analysis which is more adapted to the types of data available in geography. (Spectral analysis was developed for situations in which the data are available at uniform intervals of time or space. Human society is not organized into such regular units, but generally is arranged into nested hierarchies. Examples of such systems include governmental units; townships, counties, states, and nations; or census tracts, or postal zones, or congressional districts, court jurisdictions, the system of central places, and many others. It is common to have data aggregated in terms of these types of spatial units [3]. If one is interested in examining the world by measuring scale dependent phenomena, when the data are organized into such units, then it seems reasonable to let the hierarchical system define the scales. In this paper we propose to use analysis of variance techniques to examine scale effects in hierarchical geographical data structures. We will assume throughout that the hierarchical structure has already been defined and we do not here seek to improve the hierarchy [15].

THE METHOD

To illustrate the method we consider first a non-geographical case. Academic salaries are known to vary but it is not known whether these variations are due to differences among individuals, among departments, among colleges, or among the universities. If we know all the salaries and where each individual is in the hierarchy, then we can calculate a measure of diversity of salaries for each level of aggregation. This must of course be discounted for the degrees of freedom available at each level. The result of such a computation should suggest the extent to which each level in the hierarchy participates as a contributor to variations in salary. Here the particular measure of variation does not seem critical. We have used the conventional variance, but information theoretic measures might just as readily be used. For convenience in the ensuing equations we have assumed a complete enumeration, rather than a sampling process. This is reasonable for the data which we have actually analyzed. Readers familiar with the details of the analysis of variance will recognize the fully nested fixed effects model, type I, in our procedure [2, 7, 20, 21, 25].

We now turn to geographical examples. Suppose one wishes to examine the funds expended on road maintenance in the United States, and that these data are available by the county in which these funds are spent. The county funds can be aggregated to the state level, and these to the census region level, and then finally to a national total. An important geometric attribute of this geographical hierarchy is that the physical area, e.g., in square miles, of the units at one level must be larger than the individual units which it contains at the lower level. In the academic example, a college could not have fewer individuals than a department which it contains, although it is possible for a college to be smaller than departments in an-

other college. The geographical hierarchy thus orders the levels by areal size, and this can be taken as a surrogate for scale or resolution. Analyzing the data at different levels of the hierarchy is thus equivalent to analyzing the data at different geographical scales. The average area of United States counties is approximately that of a circle of 15 mile radius, so that using county data is equivalent to a sampling of roughly every 30 miles. The sampling theorem thus suggests that wavelengths of less than 60 miles cannot be resolved. Thus it may be possible to attach a ratio meaning to the ordinal scale defined by the hierarchy. But this should not be taken too literally for it assumes that all the units at each level of the hierarchy are roughly of the same geographical size, shape, and orientation. This is perhaps true in a central place hierarchy, but "national scale" may have little or no *spatial* meaning because the variation in the sizes of countries is too great. Lichtenstein cannot really be compared with the Soviet Union in terms of spatial scale. Of course a "county scale" or a "state scale" has a reasonable interpretation for the example at hand, because a county or state does have a jurisdictional impact on road maintenance: there are county roads, state roads, and federal roads (these are connected to each other in a hierarchical manner but they are not nested). The proportion of effort at each level must still be interpreted with caution since one would expect that climatic and population variations also affect road maintenance costs. Methods for eliminating such effects go beyond the scope of this paper, though detrending would be an obvious recommendation [11]. The point that one must have a substantive interpretation before an analysis procedure is useful [22]. Our example is somewhat contrived since the county, state, and federal funds are usually separated at the accounting level, but this would not always be the case. Thus, if one has an interpretative model, then our method will allow the inquirer to assign variances to the county, state, or regional levels of the hierarchy. Such measures may give clues to the scale at which the phenomena of concern are operating. The method then suggests "where the action is," or where it is not.

In terms of the foregoing hypothetical example, with four hierarchical levels, the model is simply:

$$X_{ijk} = \mu + \alpha_i + \beta_{ij} + \gamma_{ijk} \quad (1)$$

where X_{ijk} is the value of the k^{th} county of the j^{th} state of the i^{th} region of the county:

- μ is the grand mean for the county
- α_i is the effect of the i^{th} region
- β_{ij} is the effect of the ij^{th} state
- γ_{ijk} is the effect of the ijk^{th} county.

In contrast to an ordinary analysis of variance model there is no error term for, except in very extraordinary circumstances, multiple trials are

not performed; we have assumed a complete enumeration and thus the census-type information is taken to be correct. In contrast to general usage in the fixed effects model, the calculation of the individual fixed effects themselves is replaced by the calculation of the variance of these effects. Thus it is the variance which corresponds to the diversity at any of the given scale levels.

The grand mean and the effects may now be defined in terms of X such that

$$\mu = \frac{\sum_i \sum_j \sum_k X_{ijk}}{N} = \bar{X}_{...} \quad (2)$$

where N equals the total number of X_{ijk} observations and $\bar{X}_{...}$ indicates that this is the mean of X summed over the subscripts which are dots. Analysis of variance notation frequently includes this as a convenience. Further

$$\alpha_i = \bar{X}_{i..} - \bar{X}_{...} \quad \text{where } \bar{X}_{i..} = \frac{\sum_j \sum_k X_{ijk}}{n_i} \quad (3)$$

and n_i is the number of areas in the i^{th} region,

$$\beta_{ij} = \bar{X}_{ij.} - \bar{X}_{i..} \quad \text{where } \bar{X}_{ij.} = \frac{\sum_k X_{ijk}}{n_{ij}} \quad (4)$$

and n_{ij} is the number of areas in the j^{th} state of the i^{th} region,

$$\gamma_{ijk} = X_{ijk} - \bar{X}_{ij.} \quad (5)$$

It is clear that

$$N = \sum_i n_i = \sum_i \sum_j n_{ij} \quad (6)$$

Equation (1) may be reworked to form

$$X_{ijk} - \mu = \alpha_i + \beta_{ij} + \gamma_{ijk} \quad (7)$$

Substituting for α , β , γ , and μ one gets

$$X_{ijk} - \bar{X}_{...} = (\bar{X}_{i..} - \bar{X}_{...}) + (\bar{X}_{ij.} - \bar{X}_{i..}) + (X_{ijk} - \bar{X}_{ij.}) \quad (8)$$

Squaring

$$(X_{ijk} - \bar{X}_{...})^2 = (\bar{X}_{i..} - \bar{X}_{...})^2 + (\bar{X}_{.ij} - \bar{X}_{...})^2 + (X_{ijk} - \bar{X}_{.ij})^2 \quad (9)$$

in which the covariance terms drop out. Summing over all subscripts:

$$\begin{aligned} \sum_{i=1}^I \sum_{j=1}^{J_i} \sum_{k=1}^{K_{ij}} (X_{ijk} - \bar{X}_{...})^2 &= \sum_{i=1}^I \sum_{j=1}^{J_i} \sum_{k=1}^{K_{ij}} (\bar{X}_{i..} - \bar{X}_{...})^2 \\ &+ \sum_{i=1}^I \sum_{j=1}^{J_i} \sum_{k=1}^{K_{ij}} (\bar{X}_{.ij} - \bar{X}_{...})^2 + \sum_{i=1}^I \sum_{j=1}^{J_i} \sum_{k=1}^{K_{ij}} (X_{ijk} - \bar{X}_{.ij})^2 \quad (10) \end{aligned}$$

where I is the number of α level groups, J_i is the number of β level groups in each i^{th} α group while K_{ij} is the number of γ level groups in each ij^{th} β level group. It may be seen that:

$$\sum_i \sum_j K_{ij} = N. \quad (11)$$

Equation 10 represents the sums of squares where

$$SS_{\text{total}} = SS_{\alpha} + SS_{\beta} + SS_{\gamma} \quad (12)$$

The above equation shows how the variation around the grand mean, SS_{total} may be partitioned into parts attributable to the various scales established in the study. As in the example given at the outset of this paper, SS_{α} , SS_{β} , and SS_{γ} , are the partitions of the total variation attributable to the regional, state, and county effects.

Mean square estimates are then derived by dividing the partitioned sums of squares by the appropriate degrees of freedom, as in Table 1. Un-

TABLE 1
FORMULAE FOR THE SUMS OF SQUARES

Source of Variation	Sum of Squares	DF	Mean Square
α	$SS_{\alpha} = \sum_{i=1}^I \sum_{j=1}^{J_i} \sum_{k=1}^{K_{ij}} (\bar{X}_{i..} - \bar{X}_{...})^2$	$I - 1$	$SS_{\alpha} / I - 1$
β	$SS_{\beta} = \sum_{i=1}^I \sum_{j=1}^{J_i} \sum_{k=1}^{K_{ij}} (\bar{X}_{.ij} - \bar{X}_{...})^2$	$\sum_{i=1}^I (J_i - 1)$	$SS_{\beta} / \sum_{i=1}^I (J_i - 1)$
γ	$SS_{\gamma} = \sum_{i=1}^I \sum_{j=1}^{J_i} \sum_{k=1}^{K_{ij}} (X_{ijk} - \bar{X}_{.ij})^2$	$\sum_{i=1}^I \sum_{j=1}^{J_i} (K_{ij} - 1)$	$SS_{\gamma} / \sum_{i=1}^I \sum_{j=1}^{J_i} (K_{ij} - 1)$
Total	$SS_{\text{total}} = \sum_{i=1}^I \sum_{j=1}^{J_i} \sum_{k=1}^{K_{ij}} (X_{ijk} - \bar{X}_{...})^2$	$N - 1$	

like the usual formulation of the analysis of variance techniques, the above equations contain no error term. In census-type data multiple observations on each cell are not done and so the assumption must be made that the data are correct. (If data are obtained with multiple observations on each cell, then an additional error term may be added to control for observation error.) In such a case it might be wiser to turn to an analysis of variance type II model.

The equations presented above for calculation of the sums of squares, degrees of freedom, and the mean squares are perfectly general. However there are circumstances in which the number of areas nested at any level are a constant and in this case a simplification of the notation is possible. An example is given below using geographical data assembled by grid cells. The more realistic case is that of an uneven nesting; the number of counties for example is not the same in all states. This complicates the calculation of the actual variances of the scale components. The computation differs for the even and uneven designs and is thus discussed separately.

THE EVEN CASE

The proposed procedure is general in that it can be applied to geographical data which are regularly or irregularly spaced. The regularly spaced situation may be represented by data assembled in grid cells. Grouping of the cells is a natural way in which to form a hierarchy from such data. Discussion of such an example may make clearer just what the scale-variance components represent. Suppose one has an 8 x 8 array of data as in Figure 1, which can be totally nested. As shown in the figure, the square grid cells nest in a ratio of 4:1 from the lower level to the higher level. Such a uniform and regular nesting simplifies the algebra, i.e.,

$$J_1 = J_2 = J_3 = J_4 \quad (13)$$

$$K_{11} = K_{12} = K_{13} = \dots = K_{34} = K_{44} . \quad (14)$$

Further, in this special case

$$I = J_n = K_{nn} = 4 . \quad (15)$$

Then it becomes clear that

$$N = IJK , \quad n_i = JK \quad (16)$$

where subscripts are not needed because all J's are equal and all K's are equal while nominally

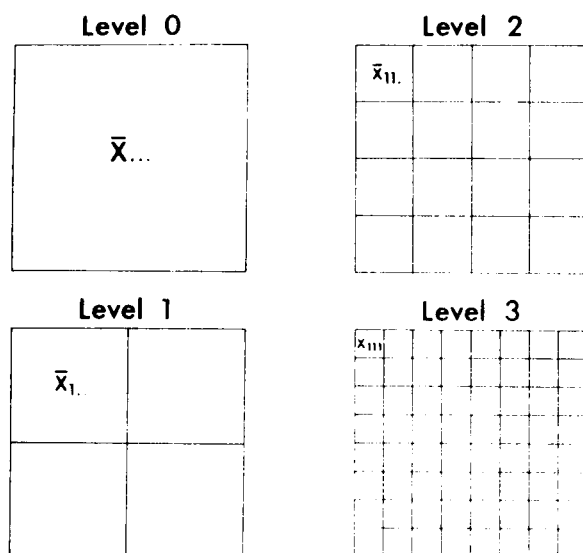


FIG. 1. An Illustration of the Nesting of an Even Hierarchy.

$$n_{ij} = K_{ij}.$$

An analysis of variance table may be constructed as above to get the expected mean squares and scale-variance parameter estimates (Table 2) in a manner similar to that of Scheffe [20]. With even numbers of cases the scale-variance components are calculated easily and simply.

TABLE 2
FORMULAE FOR THE VARIANCES: EVEN CASE

Source	Mean Square	Expected Mean Square	Scale-Variance Component
α	$MS_{\alpha} = SS_{\alpha} / I - 1$	$E(MS_{\alpha}) = \frac{JK \sum_i \alpha_i^2}{I - 1}$	$\frac{MS_{\alpha}}{JK} = \frac{\sum_i \alpha_i^2}{I - 1}$
β	$MS_{\beta} = SS_{\beta} / I(J - 1)$	$E(MS_{\beta}) = K \frac{\sum_i \sum_j \beta_{ij}^2}{I(J - 1)}$	$\frac{MS_{\beta}}{K} = \frac{\sum_i \sum_j \beta_{ij}^2}{I(J - 1)}$
γ	$MS_{\gamma} = SS_{\gamma} / IJ(K - 1)$	$E(MS_{\gamma}) = \frac{\sum_i \sum_j \sum_k \gamma_{ijk}^2}{IJ(K - 1)}$	$MS_{\gamma} = \frac{\sum_i \sum_j \sum_k \gamma_{ijk}^2}{IJ(K - 1)}$

IRREGULAR CASE

Conceptually the irregular case is the same as the gridded case. However, the calculation of the scale-variance component values is not as easy as before (Table 3). Previously, in the evenly numbered cases, the scale variance components were calculated directly from the expected mean square. Now the n_i , n_{ij} , n_{ijk} are not constants, and therefore cannot be pulled through the respective summations. Consequently, the coefficients cannot be directly calculated as before. An alternative for deriving the variances is calculating the sums of the squared deviations directly, and by-passing the mean square as shown in Table 3. If a method of calculating the coefficient directly could be found, then such a manipulation would become unnecessary.

TABLE 3
FORMULAE FOR THE VARIANCES: IRREGULAR CASE

Source	Mean Square	Expected Mean Square	Scale-Variance Components	
α	$SS_\alpha / I - 1$	$E(MS_\alpha) = \frac{\sum_i n_i \alpha_i^2}{I - 1}$	$\frac{\sum_i (\bar{X}_{i..} - \bar{X}_{...})^2}{I - 1}$	$= \frac{\sum_i \alpha_i^2}{I - 1}$
β	$SS_\beta / \sum_{i=1}^I (J_i - 1)$	$E(MS_\beta) = \frac{\sum_i \sum_j n_{ij} \beta_{ij}^2}{\sum_i (J_i - 1)}$	$\frac{\sum_i \sum_j (\bar{X}_{ij.} - \bar{X}_{i..})^2}{\sum_i (J_i - 1)}$	$= \frac{\sum_i \sum_j \beta_{ij}^2}{\sum_i (J_i - 1)}$
γ	$SS_\gamma / \sum_{i=1}^I \sum_{j=1}^{J_i} (K_{ij} - 1)$	$E(MS_\gamma) = \frac{\sum_i \sum_j \sum_k n_{ijk} \gamma_{ijk}^2}{\sum_i \sum_j (K_{ij} - 1)}$	$\frac{\sum_i \sum_j \sum_k (\bar{X}_{ijk} - \bar{X}_{ij.})^2}{\sum_i \sum_j (K_{ij} - 1)}$	$= \frac{\sum_i \sum_j \sum_k \gamma_{ijk}^2}{\sum_i \sum_j (K_{ij} - 1)}$

Unlike the previous example, the cells are not arbitrarily and uniformly grouped, but rather the cell groupings on the various scale levels have been defined in an *a priori* manner by the existence of the hierarchical system in the real world. Such a system is shown in Figure 2.

Our procedure allows the researcher to measure the scale-variance in terms of the inherent scale levels of the hierarchy itself, and thus, avoids an arbitrary grouping method by using the groupings defined by the system itself. Interpretation may then proceed on the qualities of the system itself and not on a set of units arbitrarily defined for the study.

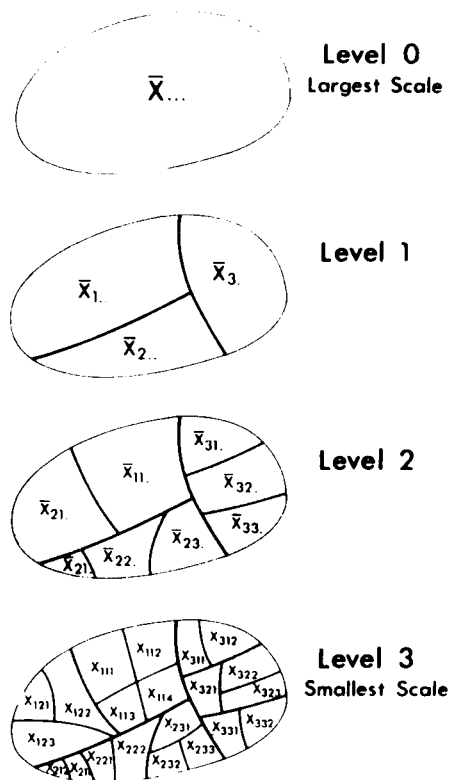


FIG. 2. An Illustration of the Nesting of an Irregular Hierarchy. The notation corresponds to that used in the text.

NUMERICAL EXAMPLES

Several numerical examples are presented here in order to illustrate the principles discussed above.

A. Regularly Spaced Data

Figure 3 shows a set of artificially constructed data to test a regular grid. The results are shown in the typical analysis of variance table as developed by R. A. Fisher (Table 4). The resulting scale-variance components may be graphed as a magnitude versus frequency graph or spectrum (Figure 4).

2	5	2	5	2	5	2	5	5	8	5	8	5	8	5	8
5	2	5	2	5	2	5	2	8	5	8	5	8	5	8	5
2	5	2	5	2	5	2	5	5	8	5	8	5	8	5	8
5	2	5	2	5	2	5	2	8	5	8	5	8	5	8	5
2	5	2	5	2	5	2	5	5	8	5	8	5	8	5	8
5	2	5	2	5	2	5	2	8	5	8	5	8	5	8	5
2	5	2	5	2	5	2	5	5	8	5	8	5	8	5	8
5	2	5	2	5	2	5	2	8	5	8	5	8	5	8	5
2	5	2	5	2	5	2	5	5	8	5	8	5	8	5	8
5	2	5	2	5	2	5	2	8	5	8	5	8	5	8	5
5	8	5	8	5	8	5	8	2	5	2	5	2	5	2	5
8	5	8	5	8	5	8	5	5	2	5	2	5	2	5	2
5	8	5	8	5	8	5	8	2	5	2	5	2	5	2	5
8	5	8	5	8	5	8	5	5	2	5	2	5	2	5	2
5	8	5	8	5	8	5	8	2	5	2	5	2	5	2	5
8	5	8	5	8	5	8	5	5	2	5	2	5	2	5	2
5	8	5	8	5	8	5	8	2	5	2	5	2	5	2	5
8	5	8	5	8	5	8	5	5	2	5	2	5	2	5	2
5	8	5	8	5	8	5	8	2	5	2	5	2	5	2	5
8	5	8	5	8	5	8	5	5	2	5	2	5	2	5	2

FIG. 3. A Numerical Example of an Even Hierarchy. The example has been artificially created by summing a high and low frequency.

TABLE 4
SCALE VARIANCES FOR THE ARTIFICIAL DATA OF FIGURE 3

Scale Level	Source of Variation	Sum of Squares	Percent of TSS	DF	Mean Square	Scale-Variance Component
4	δ_{ijk1}	576.0	50.0	192	3.0	3.0
3	γ_{ijk}	0.0	0.0	48	0.0	0.0
2	β_{ij}	0.0	0.0	12	0.0	0.0
1	α_i	576.0	50.0	3	192.0	3.0
0	Total	1152.0	100.0	255	---	---

As is clearly shown by the graph, all of the "action" is on levels 1 and 4 while there is no "action" on levels 2 and 3.

A further example uses geographical housing data given by Matui [17]. The original matrix of data was 30×40 but the last 8 columns were deleted and, using Dacey's data torus concept, the 2 rows from the top were repeated at the bottom, resulting in a 32×32 matrix of data. This is now easily nested. The results of the application of the scale-variance determination procedure are presented in Table 5. Figure 5 shows the graph of the variance component values in ascending order and shows clearly that the great-

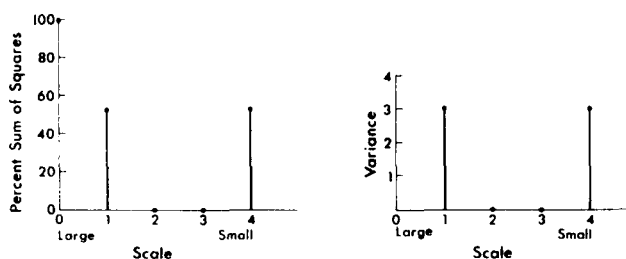


FIG. 4. The Spectrum Computed for the Data Shown in Fig. 3, Aggregated in an Even Manner to Form a Hierarchy. The spatial scales in the artificial data are perfectly recovered. Since, in the general case, the formulae (Tables 1, 2, 3) for the sums of squares for the variances differ slightly, both have been shown.

TABLE 5
SCALE VARIANCES FOR MATUI'S DATA
(THE EPSILON IS THE SMALLEST SCALE COMPONENT, NOT AN ERROR TERM)

Scale Level	Source of Variation	Sum of Squares	Percent of TSS	DF	Mean Square	Scale-Variance Component
5	ε_{ijklm}	614.25	69.37	768	.79	.79
4	δ_{ijkl}	184.8	20.98	192	.96	.24
3	γ_{ijk}	64.54	7.29	48	1.34	.0837
2	β_{ij}	15.27	1.73	12	1.27	.0199
1	α_i	5.58	0.63	3	1.86	.00727
0	Total	885.45	100.00	1024	-----	-----

est amount of the scale-variance is at level 5, the smallest cell size, and that the importance of the scale-variance components diminishes quickly as the cell sizes get larger. The interpretation is clearly that the distribution of population in Western Hukuno is attributable to local site conditions.

B. Irregularly Grouped Hierarchical Data

The 1960 Dutch census has been selected as a good example to demonstrate the methods described in this paper. The Netherlands data were selected because:

1. the hierarchy had many levels;
2. the hierarchy was fully nested;
3. the values used were totally enumerated;
4. the data were considered reliable.

The Netherlands hierarchy is generally structured into 5 levels (Figure 6) ; 980 municipalities, 78 economic geographic areas, 11 provinces, 5 regions

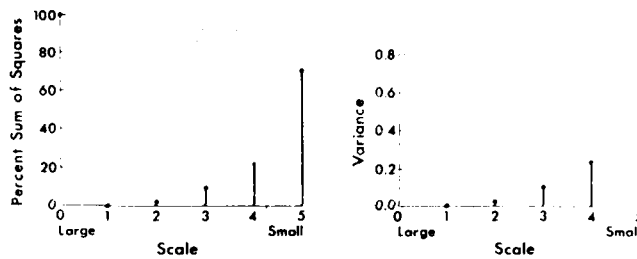


FIG. 5. The Spectrum Computed for the Matui Data, Aggregated as Described in the Text. The variance is seen to be concentrated at the smallest scales.

(not officially sanctioned, but generally accepted as valid), and finally the country as a whole. Figure 7 shows the Dutch hierarchy portrayed as a dendrogram. It is now possible to calculate the scale-variance values for the spatial units. The resolution is thus defined by the Dutch governmental hierarchy.

Scale-variance calculations for the number of inhabitants (total population) are given in Table 6. As may be seen, the bulk of the scale variation, or "action," takes place on level 3, at the economic geographic area level, with little scale variation on other levels. This may be expressed graphically in Figure 8. Notice that the hierarchical groupings are very uneven. As may be seen, the number of municipalities in an economic-geographic area is anywhere from 1 to 49. All of the larger cities, such as Amsterdam, Utrecht, Rotterdam, Haarlem, Arnhem and Eindhoven, are municipalities as well as self-contained economic areas. Conversely, the economic area of Noordelijk, Noordholland, and Westfriesland contains 49 municipalities.

Since it is known that all of the large cities are self-contained economic-geographic areas, the very large scale-variance component value on level 3 would lead one to the interpretation that the metropolitan/non-metropolitan difference is important when examining the spatial variation of population in the Netherlands, and that economic area level is the appropriate spatial scale level at which to examine the spatial variations of the pattern.

As a second example, the variable "number of males employed in agriculture" was examined (Table 7). In contrast to the preceeding example,

TABLE 6
SCALE VARIANCES FOR THE TOTAL DUTCH POPULATION (1960 CENSUS)

Scale Level	Unit	Number of Units	Sum of Squares	Percent Sum of Squares	Degree of Freedom	Mean Square Estimate	Scale Variance
4	Municipalities	980	11×10^{10}	5.98	902	13×10^7	13×10^7
3	Economic Areas	78	18×10^{11}	93.17	67	27×10^9	26×10^9
2	Provinces	11	18×10^8	0.09	6	31×10^7	43×10^5
1	Regions	5	15×10^9	0.75	4	36×10^8	25×10^6
0	Country	1	19×10^{11}	100.00	0	—	—

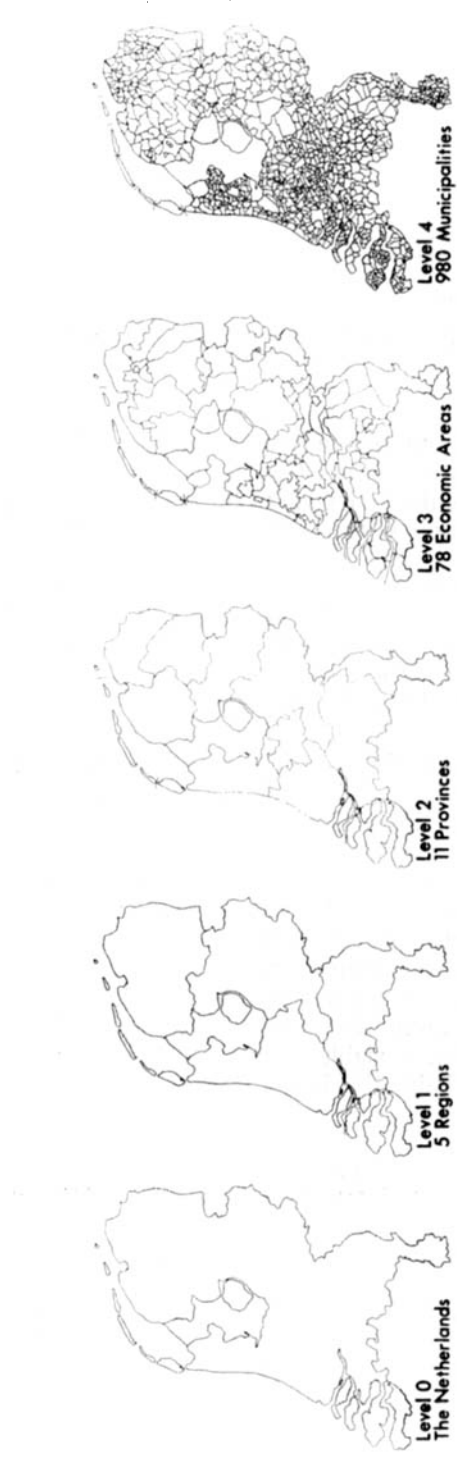


FIG. 6. Map of the Administrative Hierarchy for the Netherlands. The Municipal Level incorporates the boundary changes as of 1963. The 1960 census data used in the present analysis have been adjusted to conform to these changes.

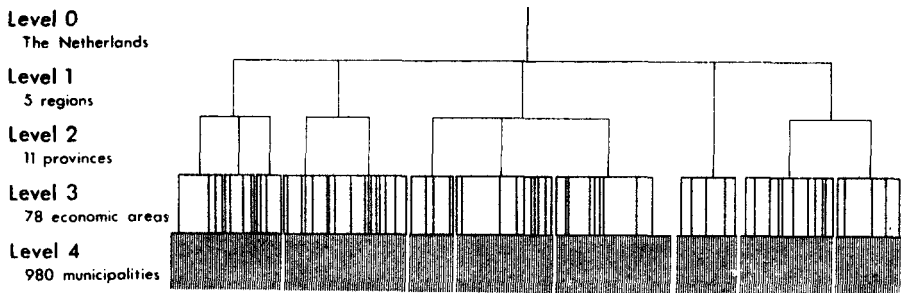


FIG. 7. Dendrogram of the Dutch Hierarchy. There 980 municipalities, 78 economic areas, 11 provinces, and 5 regions in the country. The municipalities are indicated only by shading, being somewhat too numerous to be indicated individually. Their distribution among economic areas is very uneven, ranging from one to forty-nine.

the scale-variance components are not nearly so different from one another. The spectrum of Figure 9 shows that although more action is occurring at the lower end of the spatial scale, the variance pattern of male agricultural workers is not as concentrated at scale level 3. The results may be interpreted as denoting importance in rural/urban differences on the municipal scale (level 4) as well as metropolitan/nonmetropolitan differences on the economic geographic area level. The spectrum of female agricultural workers (Table 8) indicates a somewhat greater concentration at the smaller scales.

STATISTICAL AND RELATED CONSIDERATIONS

As is clear from the preceeding examples, variance analysis provides a tool for examining the effects of spatial scales on geographically distributed data.) Comments are appropriate on several aspects of the method as now formulated. Most important is the fact that the input data must be

TABLE 7
SCALE VARIANCES FOR DUTCH MALES EMPLOYED IN AGRICULTURE (1960 CENSUS)

Scale Level	Unit	Number of Units	Sums of Squares	Percent Sum of Squares	Degrees of Freedom	Mean Square Estimate	Scale Variance
4	Municipalities	980	92×10^6	46.03	902	10×10^4	10×10^4
3	Economic Areas	78	81×10^6	40.58	67	12×10^5	50×10^4
2	Provinces	11	64×10^6	3.19	6	11×10^5	18×10^5
1	Regions	5	20×10^6	10.20	4	51×10^5	34×10^5
0	Country	1	20×10^7	100.00	0	—	—

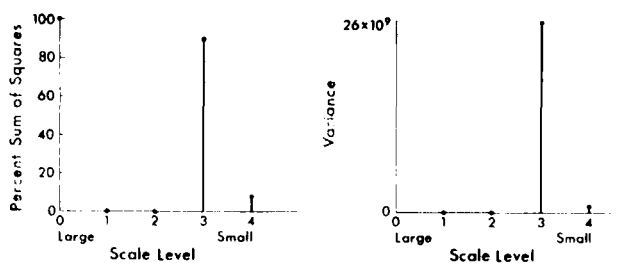


FIG. 8. The Spectrum Computed for the Distribution of Population Holland, Using the Administrative Hierarchy to Define the Spatial Resolution.

totally enumerated data and not fractionally sampled. Since this method is designed to use census and other types of geographical data where total sampling is common, the restriction is not serious.

Certain clear requirements must pertain to the hierarchy. They are three-fold:

1. The hierarchy must be fully nested, that is, that any spatial unit on a given scale level must be totally contained in one and only one unit on the spatial scale level above it (i.e., a municipality may belong to one and only one economic-geographic area, etc. see Figure 8).

2. Every branch in the hierarchy must terminate at the lowest scale level of the hierarchy. Stated differently, each spatial unit at one scale level must contain at least 1 unit of the next lower scale level.

3. "Local inversions" in the hierarchical tree are not allowed. This phenomenon was encountered by the authors while trying to find a set of numerical data for the uneven case. In most parts of the United States the spatial/political hierarchy goes: county, congressional district, state. However, in certain large metropolitan areas in the United States, the sequence is: congressional district, county, state, etc. This is illustrated in Figure 10. Such aberrations of the hierarchy, "local inversions," are clearly not allowed

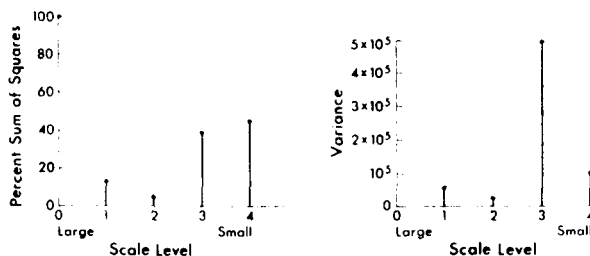


FIG. 9. The Spectrum Computed for the Number of Males Employed in Agriculture, Using the Dutch Administrative Hierarchy to Define the Spatial Resolution.

and would invalidate the sum of squares. From a geographical point of view such an occurrence would confound spatial scales.

TABLE 8
SCALE VARIANCES FOR DUTCH FEMALES EMPLOYED IN AGRICULTURE (1960 CENSUS)

Scale Level	Unit	Number of Units	Sums of Squares	Percent Sum of Squares	Degree of Freedom	Mean Square Estimate	Scale Variance
4	Municipalities	980	25×10^8	55.85	902	28×10^2	28×10^2
3	Economic Areas	78	10×10^8	22.99	67	16×10^3	21×10^2
2	Provinces	11	35×10^4	7.59	6	58×10^3	14×10^2
1	Regions	5	62×10^4	13.56	4	15×10^4	92×10^1
0	Country	1	46×10^8	100.00	0	—	—

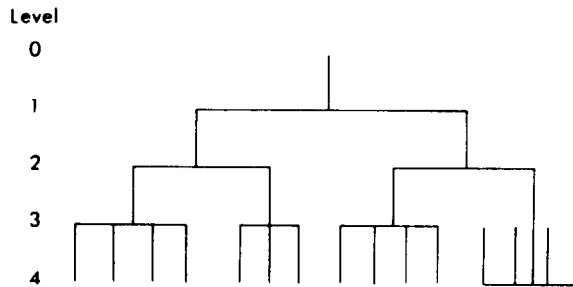


FIG. 10. Illustrating a Local Inversion in a Nested Hierarchy (See Text).

SUMMARY AND CONCLUSIONS

This paper presents a new and different method for examining geographical scale problems. As opposed to present methods of geographical scale analysis which only seem effective for data collected in particular ways, the analysis of scale variance is more flexible. The strength of our method of spatial analysis is in its ability to examine geographical scale problems in terms of the spatial units of the hierarchy available. We hope that this type of analysis will shed light on important questions concerning geographical processes. Extensions of the method to comparisons of different spatial series, collected according to different hierarchies (i.e., cross spectral analysis), or to dynamic filtering concepts, should also prove useful. Impedence mismatches between different agencies of government, for example, may be due to the differences in spatial scales of jurisdiction. Technology has rather obviously had an impact on the scales of human activity. The general trend is clear but the exact manner in which the spectra have changed over time, i.e., the transfer function, is not known in detail. We believe that the techniques described here may be helpful in elucidating such questions.

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